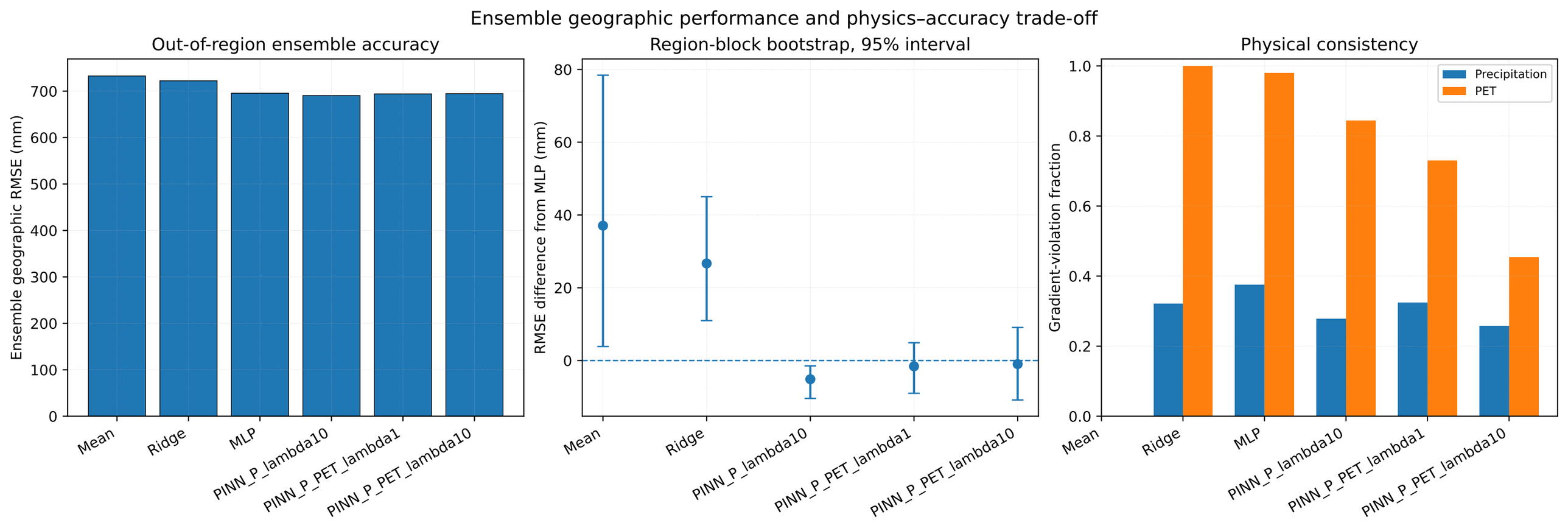
Ensemble geographic performance and physics–accuracy trade-off
Out-of-region ensemble accuracy
Ensemble geographic RMSE (mm)
0
100
200
300
400
500
600
700
Region-block bootstrap, 95% interval
RMSE difference from MLP (mm)
0
20
40
60
80
Physical consistency
Gradient-violation fraction
0.0
0.2
0.4
0.6
0.8
1.0
Precipitation
PET
Mean
Ridge
MLP
PINN_P_lambda10
PINN_P_PET_lambda1
PINN_P_PET_lambda10